

Program: Diploma in Food Processing Technology	
Course Code: 5422	Course Title: Food Additives
Semester: 5	Credits: 4
Course Category: Program core	
Periods per week: 4(L: 3T:1 P: 0)	Periods per semester: 60

Course Objectives:

1. To understand various food additives used its types, benefits, risks etc.
2. To study various colorants and flavourants used in the food industry
3. To impart knowledge on basic chemistry and mechanism of emulsifiers.

Course Prerequisites:

Topic	Program/Course Name
Basic knowledge of food additive	Food preservation

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Outline the scope, purpose, and regulation of food additives, including their types, benefits, risks, and intake assessment.	15	Understanding
CO2	Identify the properties, functions, nature, creation, production, and regulatory aspects of flavoring agents, ensuring understanding of flavor safety.	15	Understanding
CO3	Explain the chemistry, sources, applications, and safety of food colorants, including natural and synthetic types, and describe the chemistry of browning reactions and the role of anti-browning agents in food systems.	15	Understanding
CO4	Explain the basic chemistry and mechanisms of emulsifiers.	15	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Outline the scope, purpose, and regulation of food additives, including their types, benefits, risks, and intake assessment.		
M1.01	Describe on need and use of food additives	5	Understanding
M1.02	Explain types of additives	5	Understanding
M1.03	Discuss Scope and purpose of food additive	5	Understanding
Contents: Introduction to Food Additives: Definition, Types of additives, Benefits of additives, risk of additives; Scope and purpose of food additive, intake assessment, regulation of maximum levels of food additives			
CO2	Identify the properties, functions, nature, creation, production, and regulatory aspects of flavoring agents, ensuring understanding of flavor safety.		
M2.01	Describe Flavoring agents	5	Understanding
M2.02	Outline the nature of flavors and its creation and production	5	Understanding
M2.03	Discuss about flavor regulations	4	Understanding
	Series Test –I	1	

Contents: Flavoring Agent-definition, properties, functions of flavoring agents, their nature, creation and production. Regulations of flavors, flavor safety.			
CO3	Explain the chemistry, sources, applications, and safety of food colorants, including natural and synthetic types, and describe the chemistry of browning reactions and the role of anti-browning agents in food systems.		
M3.01	Discuss about Colorants	5	Understanding
M3.02	Classify different types of colorants	5	Understanding
M3.03	Explain Anti-browning agents:	5	Understanding
Contents: Colorants-chemistry, sources, effect on foods applications, safety-Types of colorants-Natural and synthetic food colorants, Anti-browning agents: Chemistry of browning reactions in foods, browning inhibitors			
CO4	Explain the basic chemistry and mechanisms of emulsifiers.		
M4.01	Discus about Emulsifiers	5	Understanding
M4.02	Explain Anti-caking agents	5	Understanding
M4.03	Explain the concept and mechanism of firming agents	4	Understanding
	Series Test–II	1	
Contents: Emulsifiers: Chemistry, function, mechanism and application; Anti-caking agents-concept, mechanism- Firming agents-concepts, mechanism			

Text / Reference

T/R	Book Title/Author
R1	Food Additives by A. Larry Branen, P.Michael Davidson, Seppo Salminen, John H. Thorngate
R2	Food Chemistry by Fennema, Food Science & technology series, 4th edition.
R3	Food Chemistry by Belitz, Grosch, Springer.

R4	Various acts, orders, standards & specification
R5	Rheology and Texture in Food Quality by J.M.DeMan
R6	Food Analysis : Theory and practice, IS: 6273 (Part-1& Part-2) by Y. Pomeranz
R7	Principles of Sensory Analysis of Food by M.A. Amerine

Online Resources

Sl.No	WebsiteLink
1	https://www.who.int/news-room/fact-sheets/detail/food-additives
2	https://www.cspinet.org/article/artificial-colorings-synthetic-food-dyes
3	https://learn.ddwcolor.com/what-are-natural-colors/
4	https://www.webmd.com/diet/what-are-emulsifiers